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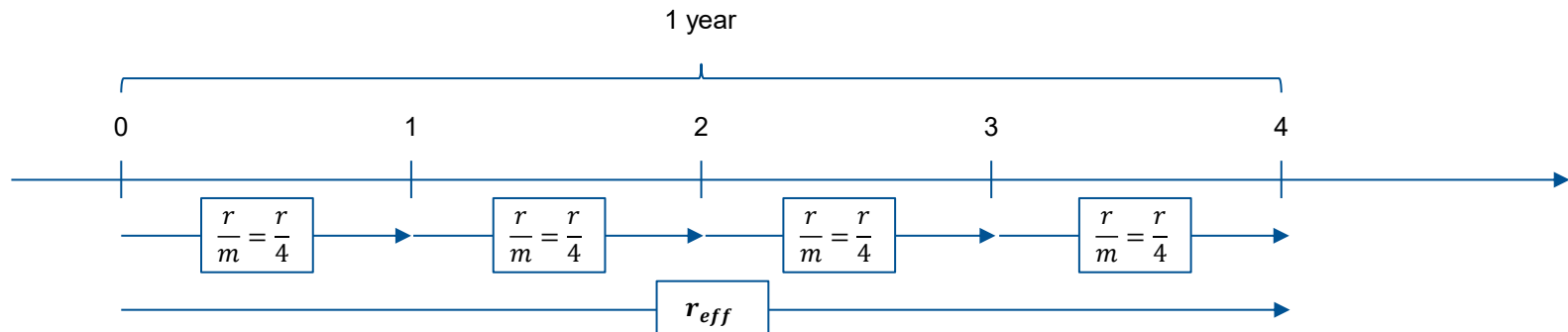
1.6 Exercises and Solutions

Compounding frequency and effective interest rate

- **Problem:** In contracts interest rates are typically stated on an annual basis. This is called the nominal interest rate (r). However, compounding often takes place on monthly or quarterly intervals.
- **Examples in Germany**
 - Current account in Germany: quarterly interest payments ($m = 4$)
 - Loan accounts in Germany: monthly interest payments ($m = 12$)
 - Saving accounts: yearly interest payments ($m = 1$)
- Every interest period (e.g. quarter or month) has the same length ($\frac{360}{m}$)

Effective interest rate with intra-year interest

- Definition:** $\frac{r}{m}$ equals the interest rate per period when r is the annual (nominal) interest rate



- If interests are charged at the end of one period, in the next period the interest rate is paid on the principal amount plus the interest charged for the preceding period (compounding effect). So, ...
 - r equals the nominal annual interest rate, and
 - r_{eff} equals the effective annual interest rate.

$$(1): r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1$$

Derivation of the effective interest rate:

$$(1): C_0 * \left(1 + \frac{r}{m}\right)^m = C_0 * (1 + r_{eff})$$

$$(2): \left(1 + \frac{r}{m}\right)^m = (1 + r_{eff})$$

$$(3): r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1$$

There are two ways of calculating the future value with intra-year interests!

1. Adjust the interest period to the interest rate

- Calculations are based on the intra-year interest rate $\frac{r}{m}$ instead of the annual interest rate
- Calculation of the interest periods N :
(1): $N = n * m \rightarrow C_N = C_0 * \left(1 + \frac{r}{m}\right)^N$, with
(2): $N = \text{Number of interest periods over the entire period}$
(3): $n = \text{Investment period in years, and } m = \text{interest periods per years}$

2. Adjust the interest rate to the interest period

- Calculation of the effective annual interest rate

$$r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1 \rightarrow C_n = C_0 * (1 + r_{eff})^n$$

(A.6a) Nominal annual interest rates are given: 2.35%, 2.00%, and 1.75%. Please calculate the relative interest rate per period and the effective interest rate if we assume quarterly interest payments.

Nominal interest rate	Rel. interest rate per period	Effective interest rate
$r = 2.35\%$	$\frac{r}{m} = \frac{2.35\%}{4} = 0.59\%$	$= (1 + \frac{0.0235}{4})^4 - 1$ $= 2.37\%$
$r = 2.00\%$	$\frac{r}{m} = \frac{2.00\%}{4} = 0.50\%$	$= (1 + \frac{0.0200}{4})^4 - 1$ $= 2.02\%$
$r = 1.75\%$	$\frac{r}{m} = \frac{1.75\%}{4} = 0.44\%$	$= (1 + \frac{0.0175}{4})^4 - 1$ $= 1.76\%$

(A.6b) Please evaluate the difference between a yearly investment of 50,000 EUR on a quarterly (1) and six-monthly (2) interest basis if you assume an identical nominal interest rate of 2.50%.

$$(F): C_n = C_0 * \left(1 + \frac{r}{m}\right)^m, \text{ with } r = 0.025$$

$$(1): C_n = 50,000 \text{ EUR} * \left(1 + \frac{0.025}{4}\right)^4 = 51,261.77 \text{ EUR}$$

$$(2): C_n = 50,000 \text{ EUR} * \left(1 + \frac{0.025}{2}\right)^2 = 51,257.81 \text{ EUR}$$

$$(3): \text{Difference} = (1) - (2) = 3.96 \text{ EUR}$$

Disclosure of effective interest rates in consumer loan contracts

- According to Art. 30, Directive EU 2023/2225 consumer loan contracts have to disclose the effective interest rate of the loan
- The way how the effective interest rate is calculated follows the ICMA rules, e.g.
 - ✓ Interest days are calculated according to the actual/actual rule
 - ✓ All payments associated with the contract have to be accounted for
- For consumer mortgage loans similar rules can be found in Art. 17, Directive 2014/17/EU
- In Germany, these rules have been transformed into national law with the Preisangabenverordnung (PangV)

(A.7) You buy a new computer at a price of 1,250 EUR. The Hell's Bank offers you a loan at the following conditions. It has to be repaid in two equal instalments of 784 EUR due after 6 and 12 months. What is the effective interest rate of this loan?

$$1,250 = \frac{784}{(1+r_{eff})^{0.5}} + \frac{784}{(1+r_{eff})^1}$$

$$r_{eff} = 35.81\%$$

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Continuous compounding interest

Definition: Interest on invested capital is calculated on a continuous basis.

- This means that interests are paid and reinvested infinite times during the investment period.
- The number of interest periods becomes infinite.

$$(1): C_n = C_0 * \left(1 + \frac{r}{m}\right)^m, \text{ with } m \rightarrow \infty$$

Effective interest rates with continuous compounding

1. Nominal annual interest rate

5%

10%

15%

2. Intra-year interest

m

Effective interest rate

Six-monthly

2

5.06%

10.25%

15.56%

Quarterly

4

5.09%

10.38%

15.87%

Monthly

12

5.12%

10.47%

16.08%

Weekly

52

5.12%

10.51%

16.16%

Daily

365

5.13%

10.52%

16.18%

3. Continuous compounding

5.13%

10.52%

16.18%

Effective interest rates with continuous compounding

The effective annual interest rate converges to a limit value x which can be determined as:

$$(1): x = \lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m - 1$$

$$(2): \text{We know: } e = \lim_{a \rightarrow \infty} \left(1 + \frac{1}{a}\right)^a = 2.7182 \dots, \text{ and define } a = \frac{m}{r} \Leftrightarrow m = a * r$$

Replacing $m = a * r$ in (1)

$$(3): x = \lim_{a * r \rightarrow \infty} \left(1 + \frac{1}{a}\right)^{a * r} - 1, \text{ with } r = \text{const.}$$

$$(4): x = \lim_{a \rightarrow \infty} \left[\left(1 + \frac{1}{a}\right)^a \right]^r - 1 = e^r - 1$$

$$(5): x = e^r - 1$$

If there are m subperiods during the year, then the final value for an investment over n years equals:

$$(1): C_n = C_0 * \left(1 + \frac{r}{m}\right)^{n*m} = C_0 * \left[\left(1 + \frac{r}{m}\right)^m\right]^n$$

If the value for m – the number of subperiods – becomes large, this results in a strong compounding effect:

$$(2): C_n = \lim_{m \rightarrow \infty} C_0 * \left[\left(1 + \frac{r}{m}\right)^m\right]^n = C_0 * e^{r*n}$$

For continuous compounding the effective annual interest rates with nominal interest rate r equals:

$$(3): r_{eff} = (e^r - 1)$$

(A.9) Please calculate the effective interest rate if the nominal annual rate equals 6%.

$$(F): r_{eff} = (e^r - 1)$$

$$(1): r_{eff} = (e^{0.06} - 1) = 6.18\%$$

(A.10) You have 3,000 EUR to invest for two years. Please consider two investment opportunities. Which one is better?

- a. Nominal annual interest rate of 5% with compounding interest
- b. Continuous compounding interest with 4% interest rate

A: $a > b$

B: $a = b$

C: $a < b$

D: *cannot be answered*

$$(F): C_2^i = C_0 * (1 + r)^n = C_0 * q^n$$

$$(1): C_2^a = 3,000 \text{ EUR} * 1.05^2 = 3.307,50 \text{ EUR}$$

$$(2): C_2^b = C_0 * e^{r*n} = 3,000 \text{ EUR} * e^{0.04*2} = 3,249.86 \text{ EUR}$$

$$(3): C_2^a > C_2^b, \Delta = 57.64 \text{ EUR}$$

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(A.11) At which interest rate do you invest 3,000 EUR to be able to receive a cash flow of 5,000 EUR after 12 years if you assume simple interests.

$$(F): C_n = C_0 * (1 + n * r) \leftrightarrow \frac{C_n}{C_0} = (1 + n * r)$$

$$(1): \frac{5,000 \text{ EUR}}{3,000 \text{ EUR}} = (1 + 12 * r) \leftrightarrow r = 5.56\%$$

(A.12) Let's assume King Charles I (Karl der Große) put 1 EUR on a bank account at the Kreissparkasse in Aachen in 800 when he was crowned Emperor of the Romans. How much money would be on the bank account (constant interest rate at 1% p.a.) in 2017 if interests would have been calculated on ...

- a. a simple interest basis?
- b. a compounding interest basis?

How large would the difference between the two methods be?

$$A = 0.00 \text{ EUR}$$

$$B = 1,217.00 \text{ EUR}$$

$$C = 81,321.11 \text{ EUR}$$

$$D = 181,585.18 \text{ EUR}$$

$$(1): C_{2017}^a = C_0 * (1 + n * r) = 1 \text{ EUR} * (1 + 1,217 * 0.01) = 13.17 \text{ EUR}$$

$$(2): C_{2017}^b = C_0 * (1 + r)^n = 1 \text{ EUR} * (1 + 0.01)^{1,217} = 181,598.35 \text{ EUR}$$

$$(3): C_{2017}^b - C_{2017}^a = 181,585.18 \text{ EUR}$$

(A.13): Simple Interest vs. Compounding Interest

(A.13) Consider a saving account with an interest rate of 10% p.a. and an investment of 1,000 EUR at t_0 . How long does it take until the account balance equals 5,000 EUR?

- Please assume simple interest.
- Please assume compounding interest.

$$(1): C_n^a = C_0 * (1 + n * r) \Leftrightarrow n = \frac{\frac{C_n^a}{C_0} - 1}{r} = \frac{\frac{5,000 \text{ EUR}}{1,000 \text{ EUR}} - 1}{0.10} = 40 \text{ years}$$

$$(2): C_n^b = C_0 * (1 + r)^n \Leftrightarrow n = \frac{\ln\left(\frac{C_n^b}{C_0}\right)}{\ln(1 + r)} = \frac{\ln\left(\frac{5,000 \text{ EUR}}{1,000 \text{ EUR}}\right)}{\ln(1 + 0.10)} = 16.89 \text{ years}$$

(A.14) Consider a saving account with an investment of 1,000 EUR at t_0 . Please determine the annual interest rate so that the account balance will be 1,500 EUR after 2.5 years.

- Please assume simple interest.
- Please assume compounding interest.

$$(1): C_{2.5}^a = C_0 * (1 + n * r) \leftrightarrow r = \frac{\frac{C_n^a}{C_0} - 1}{n} = \frac{\frac{1,500 \text{ EUR}}{1,000 \text{ EUR}} - 1}{2.5} = 20.00\%$$

$$(2): C_{2.5}^b = C_0 * (1 + r)^n \leftrightarrow r = \sqrt[n]{\frac{C_n}{C_0}} - 1 = \sqrt[2.5]{\frac{1,500 \text{ EUR}}{1,000 \text{ EUR}}} - 1 = 17.61\%$$

(A.15): Intra-year Interest

(A.15a) Consider an investment with monthly interests of 1%. What are the nominal annual interest rate and the effective interest rate?

$$A: r = 12.00\%, \\ r_{eff} = 12.68\%$$

$$B: r = 12.00\%, \\ r_{eff} = 12.00\%$$

$$C: r = 12.68\%, \\ r_{eff} = 12.00\%$$

$$D: r = 12.68\%, \\ r_{eff} = 12.68\%$$

(1): *Nominal annaul rate*: $\frac{r}{m} = 0.01$, with $m = 12 \rightarrow r = 12 * 0.01 = 12\%$

(2): *Eff. interest rate*: $r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1 = (1 + 0.01)^{12} - 1 = 12.68\%$

(A.15b) What six-monthly interest rate needs to be chosen in order to achieve an effective interest rate of 9%?

(1): *Eff. interest rate*: $r_{eff} = 0.09 = \left(1 + \frac{r}{2}\right)^2 - 1$

(2): $\frac{r}{2} = (\sqrt[2]{1.09} - 1) \Leftrightarrow \frac{r}{2} = 4.40\%$

(A.16) You have the opportunity to put money in an account earning 0.35% interest per month. In 18 months, you will need 1,000 EUR. How much money do you need to put on the account?

Two Solutions:

1. Monthly interest and adjustment of interest periods

$$(1): m = 12, n = 1.5$$

$$(2): C_n = C_0 * (1 + 0.0035)^{12*1.5}$$

$$(3): C_0 = \frac{1,000 \text{ EUR}}{(1.0035)^{12*1.5}} = 939.05 \text{ EUR}$$

2. Effective annual interest rate

$$(4): r_{eff} = (1 + 0.0035)^{12} - 1 = 4.28\%$$

$$(5): C_0 = \frac{1,000 \text{ EUR}}{(1 + 0.0428)^{1.5}} = 939.05 \text{ EUR}$$

(A.17) Interests are paid on an initial investment of 4,000 EUR with a nominal annual interest rate of 6%. What will be the final value in three years if interests are calculated (a) yearly, (b) six-monthly, and (c) on a continuous basis?

(a. 1): *Yearly*: $C_n = C_0 * (1 + r)^n$

(a. 2): $C_n = 4,000 \text{ EUR} * 1.06^3 = 4,764.06 \text{ EUR}$

(b. 1): *Six – month*: $C_n = C_0 * \left(1 + \frac{r}{m}\right)^{m*n}$

(b. 2): $C_n = 4,000 \text{ EUR} * \left(1 + \frac{0.06}{2}\right)^{2*3} = 4,776.21 \text{ EUR}$

(c. 1): *Continuous*: $C_n = C_0 * e^{r*n}$

(c. 2) $C_n = 4,000 \text{ EUR} * e^{0.06*3} = 4,788.87 \text{ EUR}$

(A.18): Continuous Compounding Interest

(A.18) Interest are paid on an initial investment C_0 . In t_1 the account balance increases to $C_1 = 100,000 \text{ EUR}$. After two years, the balance reaches $C_2 = 110,000 \text{ EUR}$. How much is the initial investment C_0 and the continuous compounding interest rate r ?

$$(1): C_n = C_0 * e^{r*n} \rightarrow C_1 = C_0 * e^r, \text{ and } C_2 = C_0 * e^{2*r}$$

Now we can use the capital growth from year 1 to year 2:

$$(2): \frac{C_2}{C_1} = \frac{C_0 * e^{2*r}}{C_0 * e^r} = e^r \Leftrightarrow r = \ln\left(\frac{C_2}{C_1}\right)$$

$$(3): r = \ln\left(\frac{110,000 \text{ EUR}}{100,000 \text{ EUR}}\right) = 9.53\%$$

$$(4): C_0 = \frac{C_1}{e^r} = \frac{100,000 \text{ EUR}}{e^{0.0953}} = 90,909.09 \text{ EUR}$$

Summary: Formulas

1. Future Value

$$C_n = C_0 * (1 + n * r)$$

2. Present Value

$$C_0 = \frac{C_n}{(1 + n * r)}$$

3. Time

$$C_n = C_0 * (1 + n * r)$$

$$n = \frac{\frac{C_n}{C_0} - 1}{r}$$

4. Interest Rate

$$C_n = C_0 * (1 + n * r)$$

$$r = \frac{\frac{C_n}{C_0} - 1}{n}$$

1. Future Value

$$C_n = C_0 * (1 + r)^n = C_0 * q^n$$

2. Present Value

$$C_0 = \frac{C_n}{(1 + r)^n} = \frac{C_n}{q^n}$$

3. Time

$$q^n = \frac{C_n}{C_0}; \ln(q^n) = \ln\left(\frac{C_n}{C_0}\right)$$

$$n * \ln(q) = \ln(C_n) - \ln(C_0)$$

$$n = \frac{\ln(C_n) - \ln(C_0)}{\ln(q)}$$

4. Interest Rate

$$q = \sqrt[n]{\frac{C_n}{C_0}}$$

$$r = \sqrt[n]{\frac{C_n}{C_0}} - 1$$

1. Future Value

$$C_n = C_0 * e^{r*n}$$

2. Present Value

$$C_0 = \frac{C_n}{e^{r*n}}$$

3. Time

$$C_n = C_0 * e^{r*n}$$

$$e^{r*n} = \frac{C_n}{C_0} \leftrightarrow r * n = \ln\left(\frac{C_n}{C_0}\right)$$

$$n = \frac{\ln(C_n) - \ln(C_0)}{r}$$

4. Interest Rate

$$C_n = C_0 * e^{r*n}$$

$$e^{r*n} = \frac{C_n}{C_0} \leftrightarrow r * n = \ln\left(\frac{C_n}{C_0}\right)$$

$$r = \frac{\ln(C_n) - \ln(C_0)}{n}$$